中国人民大學 RENMIN UNIVERSITY OF CHINA

中国人民大学信息学院 SCHOOL OF INFORMATION RENMIN UNIVERSITY OF CHINA

蚂蚁集团 ANT GROUP

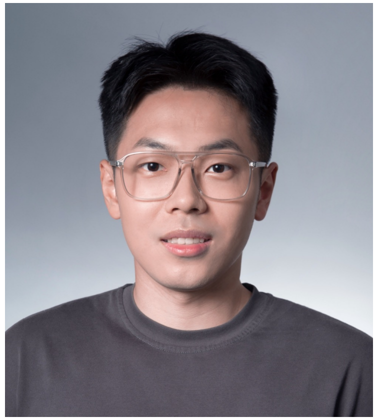
DeepAnalyze:
Agentic自主数据科学大模型

张绍磊

中国人民大学

2026.03.25

个人基本情况



张绍磊 / 中国人民大学信息学院 助理教授

研究方向

- 人工智能（大模型、多模态、Agentic）
- 数据科学（数据治理、数据分析）

大模型项目在GitHub社区获得广泛关注，累积获得 10000+ 星标

ictnlp/BayLing Public
"百聆"是一个基于LLaMA的语言对齐增强的英语/中文大语言模型，具有优越的英语/中文能力，在多语言和通用任务等多项测试中取得ChatGPT 90%的性能。BayLing is an English/Chinese LLM equipped with advanced language alignment, showing superior capability in English/Ch...
Python 318 20

BayLing 318 Stars

ictnlp/StreamSpeech Public
StreamSpeech is an "All in One" seamless model for offline and simultaneous speech recognition, speech translation and speech synthesis.
Python 1.2k 98

StreamSpeech 1200+ Stars

LLaMA-Omni Public
LLaMA-Omni is a low-latency and high-quality end-to-end speech interaction model built upon Llama-3.1-8B-Instruct, aiming to achieve speech capabilities at the GPT-4o level.
Python 3.1k 216

LLaMA-Omni 3100+ Stars

LLaVA-Mini Public
LLaVA-Mini is a unified large multimodal model (LMM) that can support the understanding of images, high-resolution images, and videos in an efficient manner.
Python 543 28

LLaVA-Mini 548 Stars

ictnlp/Stream-Omni Public
Stream-Omni is a GPT-4o-like language-vision-speech chatbot that simultaneously supports interaction across various modality combinations.
Python 355 38

Stream-Omni 385 Stars

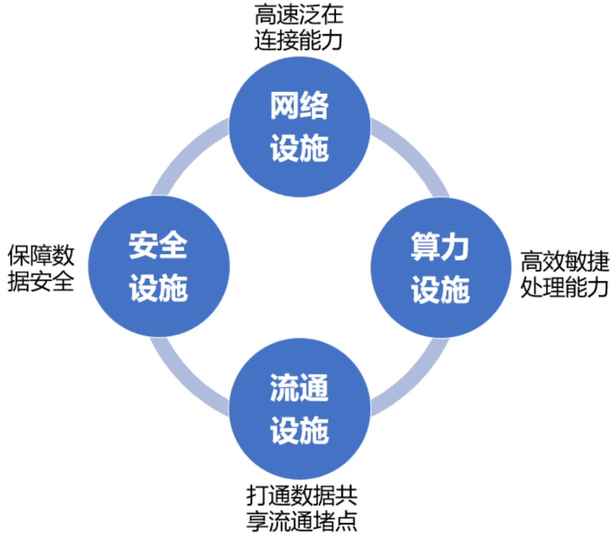
ruc-datalab/DeepAnalyze Public
DeepAnalyze is the first agentic LLM for autonomous data science. 你的AI数据分析师，自动分析大量数据，一键生成专业分析报告！
Python 2.3k 325

DeepAnalyze 3800+ Stars

数据科学是释放数据价值的关键

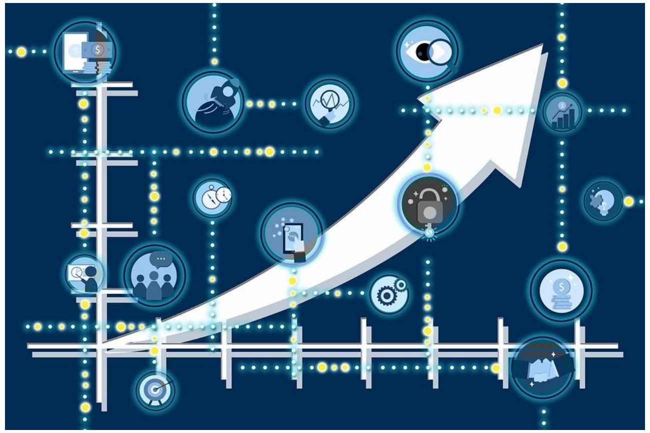
数据科学 提升数据科学系统的智能化水平与性能，高效释放数据价值

助力数据基础设施建设



促进智慧城市精细治理

推动产业数智化转型



数据科学技术起到了重要的基础性支撑作用

来源：《国家数据局数据技术设施体系》《数据空间发展战略蓝皮书》

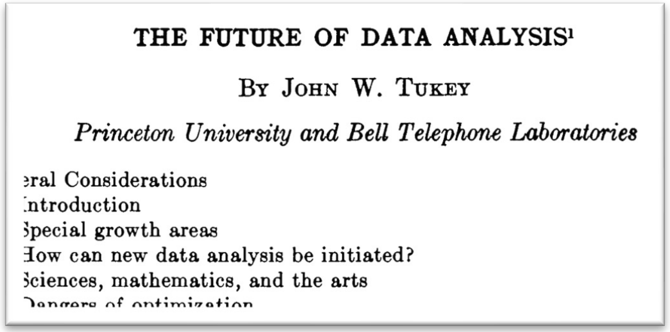
什么是数据科学？

数据技术 ＋ 智能技术

数据技术的发展历程

数据技术　从数据库技术到大数据技术

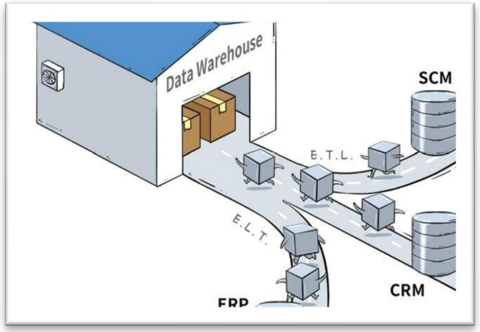
数据来源　数据规模　数据质量


THE FUTURE OF DATA ANALYSIS[1]

By John W. Tukey

Princeton University and Bell Telephone Laboratories

eral Considerations
ntroduction
Special growth areas
How can new data analysis be initiated?
Sciences, mathematics, and the arts
Dangers of optimization

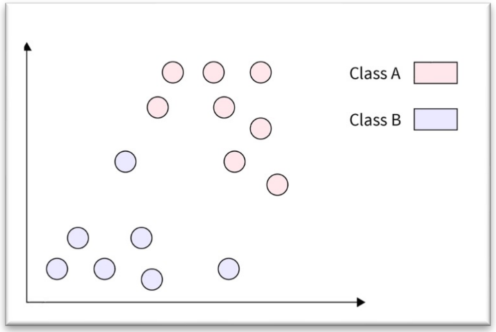


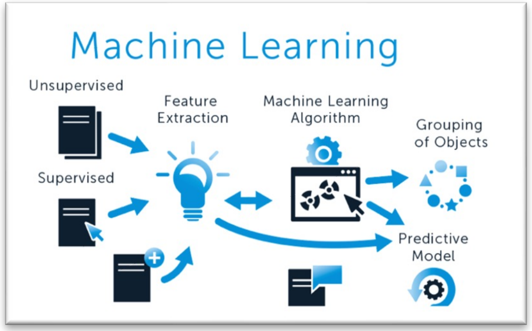
数据科学概念被提出	关系数据库、数据仓库	数据挖掘	大数据
1962	1960s-1990s	2000s	2010s

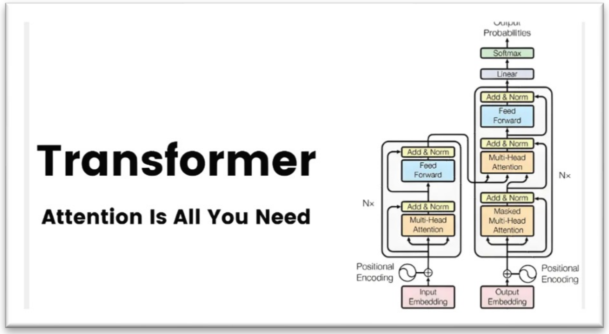
智能技术的发展历程

智能技术　从机器学习到深度学习

模态数量 ↑　模型能力 ↑　任务类型 ↑







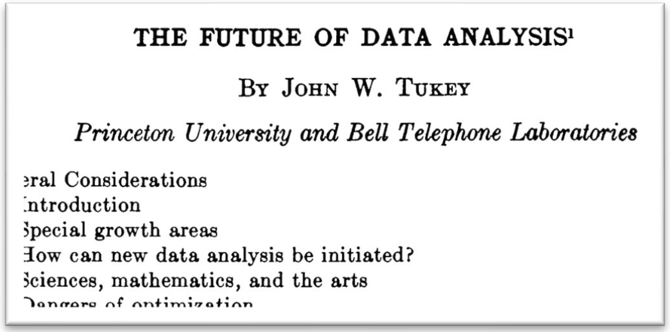
人工智能概念被提出 — 1956

专家系统、统计学习 — 1960s-1990s

机器学习 — 2000s

深度学习 — 2010s

数据技术的发展历程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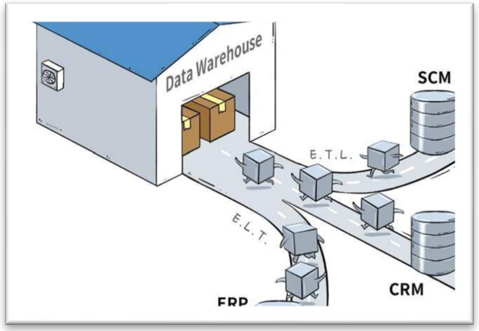
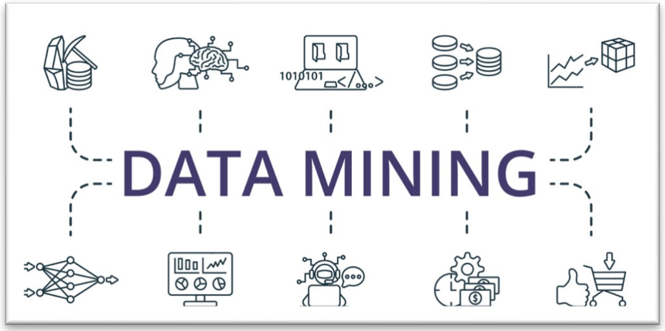
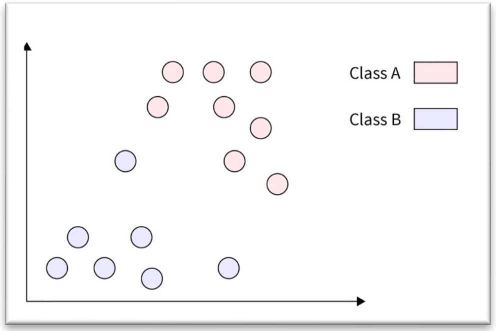


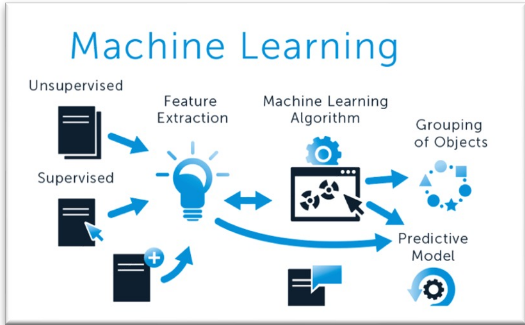


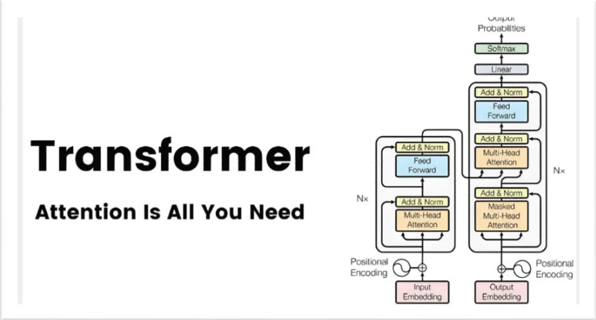
数据科学概念被提出 | 关系数据库、数据仓库 | 数据挖掘 | 大数据

~1960 | 1960s-1990s | 2000s | 2010s

人工智能概念被提出 | 专家系统、统计学习 | 机器学习 | 深度学习





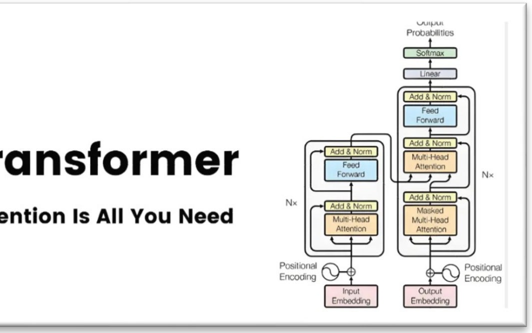


数据 + 智能

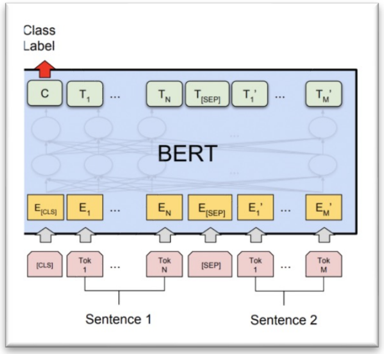
大数据

2010s

深度学习



完成数据准备、分析等少量数据科学任务

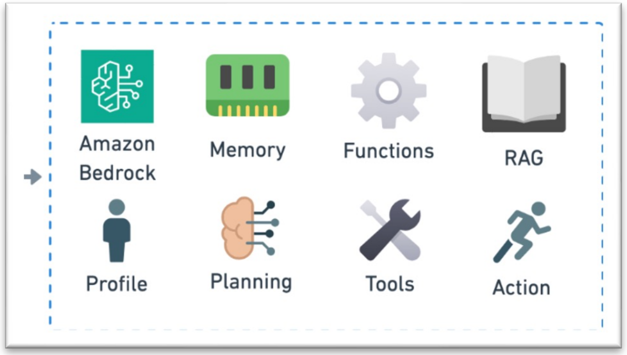


2019
BERT/GPT等
预训练模型出现

通过Prompt工程完成单一数据科学任务

2022
ChatGPT等
大模型（LLM）出现

通过Agent完成复杂数据科学Pipeline

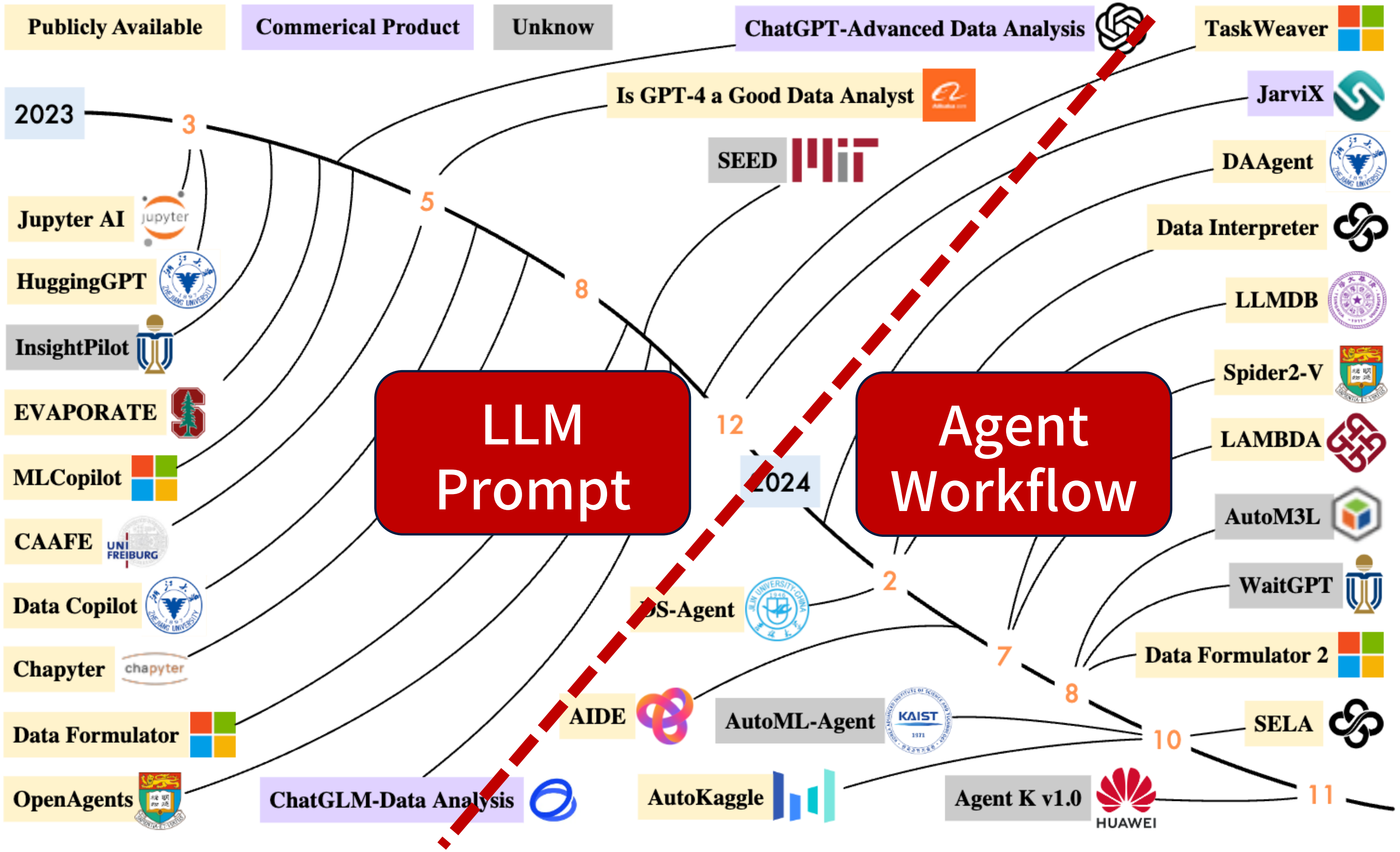


2024
Agent
智能体出现

数据科学智能体

AI for Data Science: 基于大模型的数据科学系统

以大模型为核心的数据科学系统成为研究热点



为什么大模型完成数据科学任务成为趋势？

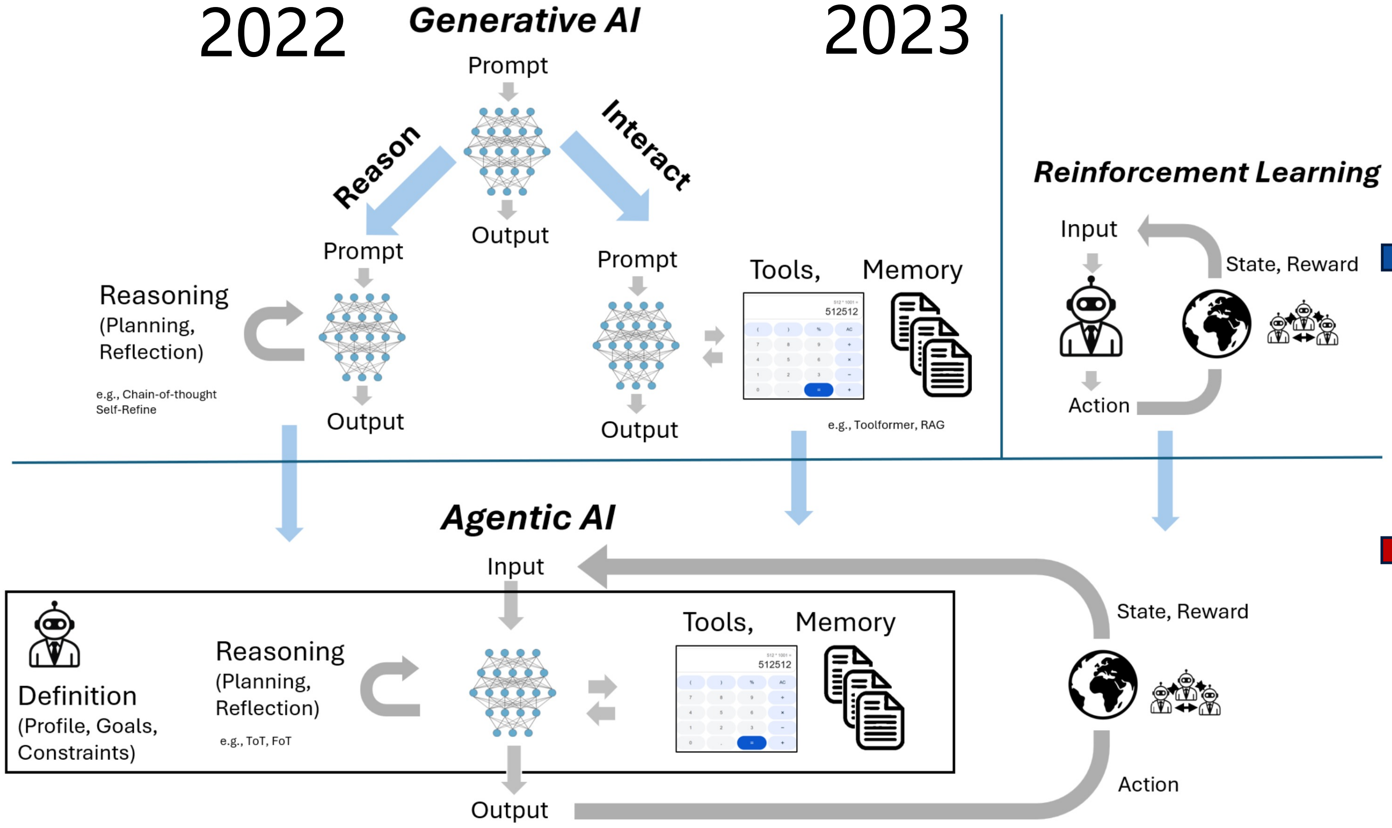
- 数据科学任务在各领域需求丰富，亟需解决
- 数据科学任务难，直接反应LLM系统的智能化水平

图源: A Survey on Large Language Model-based Agents for Statistics and Data Science

AI for Data Science: Agentic数据科学系统

Agentic AI 大模型是否能自主完成数据科学任务?

单一大模型 — 2022
Prompt Engineering — 2023
Agent Workflow — 2024
Agentic AI — 2025



生成式AI（LLM/Agent Workflow）

- 单点优化各任务（推理/执行...）
- 通过Workflow启发式组合

Agentic AI

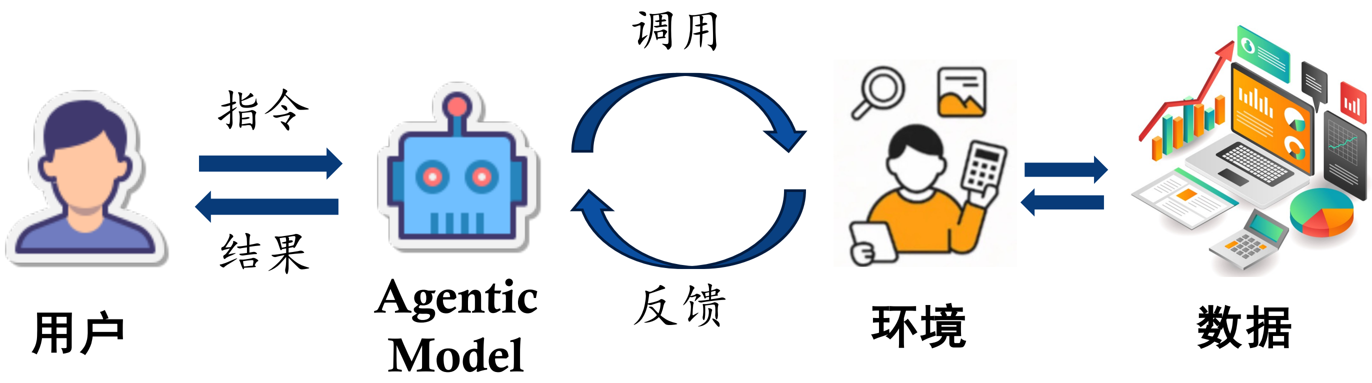
- 在真实环境中，端到端优化
- 完全自动化执行

图源: Generative to Agentic AI: Survey, Conceptualization, and Challenges

AI for Data Science: Agentic数据科学系统

Agentic AI 大模型是否能自主完成数据科学任务?

单一大模型	Prompt Engineering	Agent Workflow	Agentic AI
2022	2023	2024	2025



Agentic 数据科学系统特征:
1. 声明式交互
2. 自动化与环境交互
3. 全流程联合优化
4. 多模态感知

DeepAnalyze：自主数据科学大模型

DeepAnalyze 模拟数据科学家的行为，自动从数据源中提取insight

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Shaolei Zhang[1] **Ju Fan**[1] **Meihao Fan**[1] **Guoliang Li**[2] **Xiaoyong Du**[1]

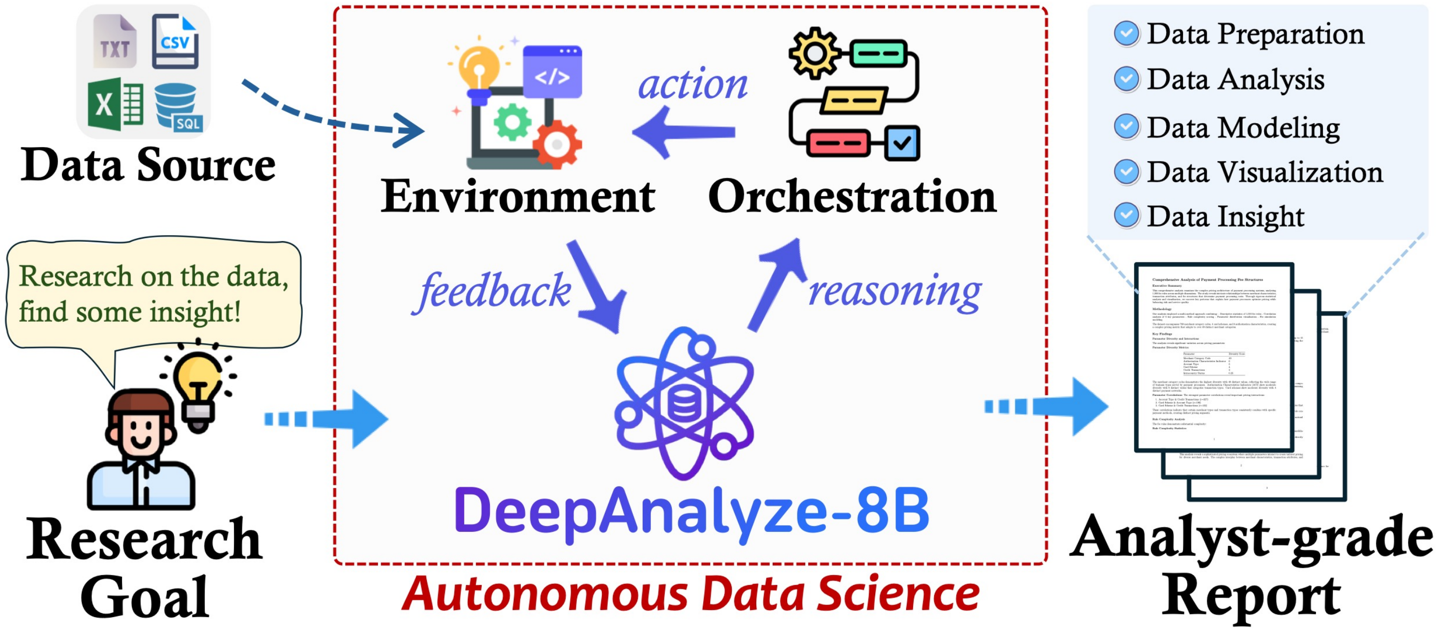
ruc-datalab/DeepAnalyze　DeepAnalyze-8B　DataScience-Instruct-500K　ruc-deepanalyze.github.io

能胜任开放域下数据科学任务：

1. 分析任务，并规划步骤
2. 在环境中读取数据
3. 观察并理解数据
4. 通过代码完成数据分析、建模、可视化...
5. 总结、呈现结果

DeepAnalyze：自主数据科学大模型

DeepAnalyze 模拟数据科学家的行为，自动从数据源中提取insight

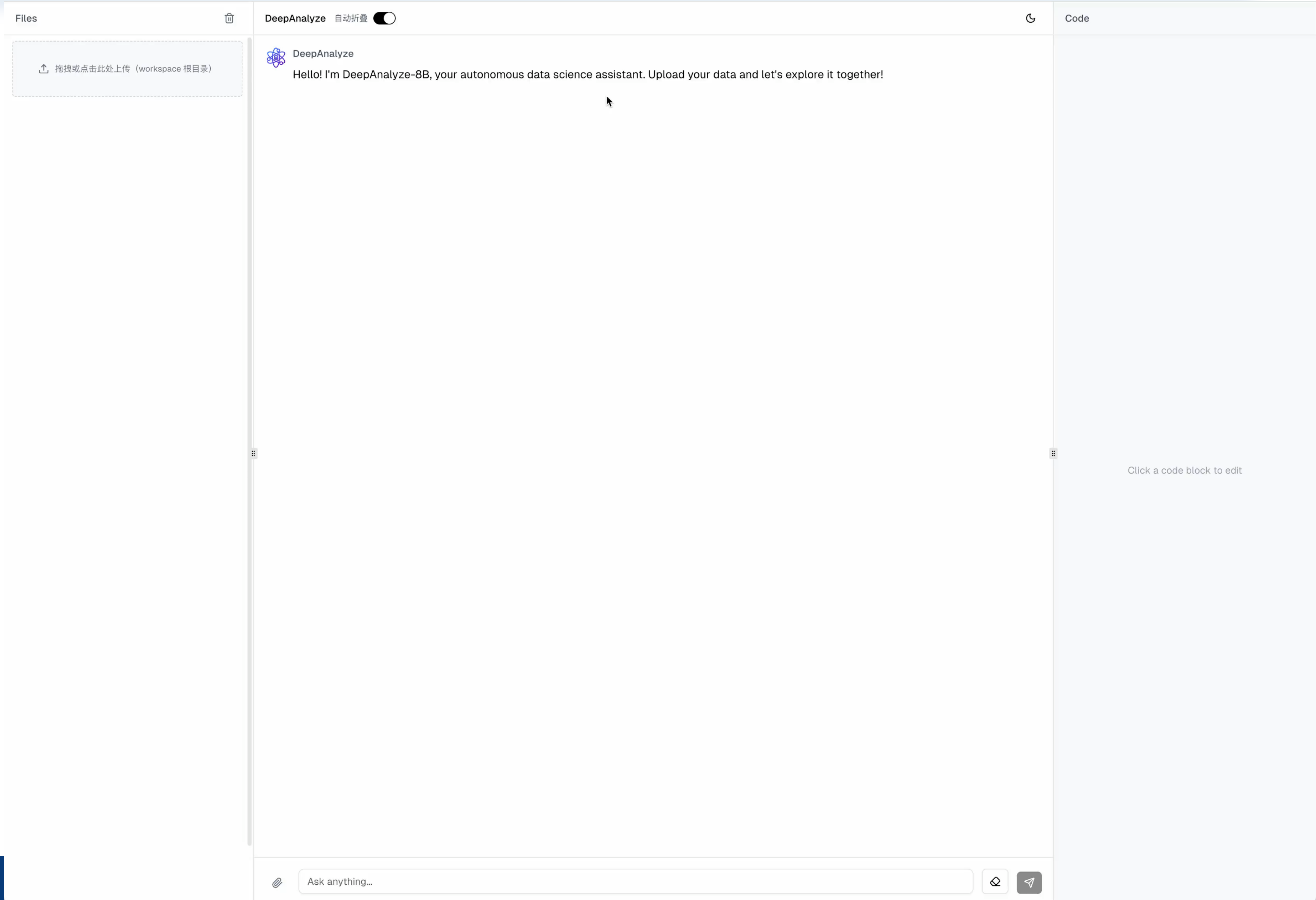


给定数据源，DeepAnalyze自主编排操作、根据反馈自适应优化

- 全流程数据科学Pipeline：数据准备、分析、建模、可视化、洞察
- 开放式数据研究：自动研究数据，生成分析师级别报告

DeepAnalyze：自主数据科学大模型

DeepAnalyze-8B
（1）处理多种类型数据
（2）自动编排+自适应优化
（3）开放式数据研究，生成分析师级别研究报告

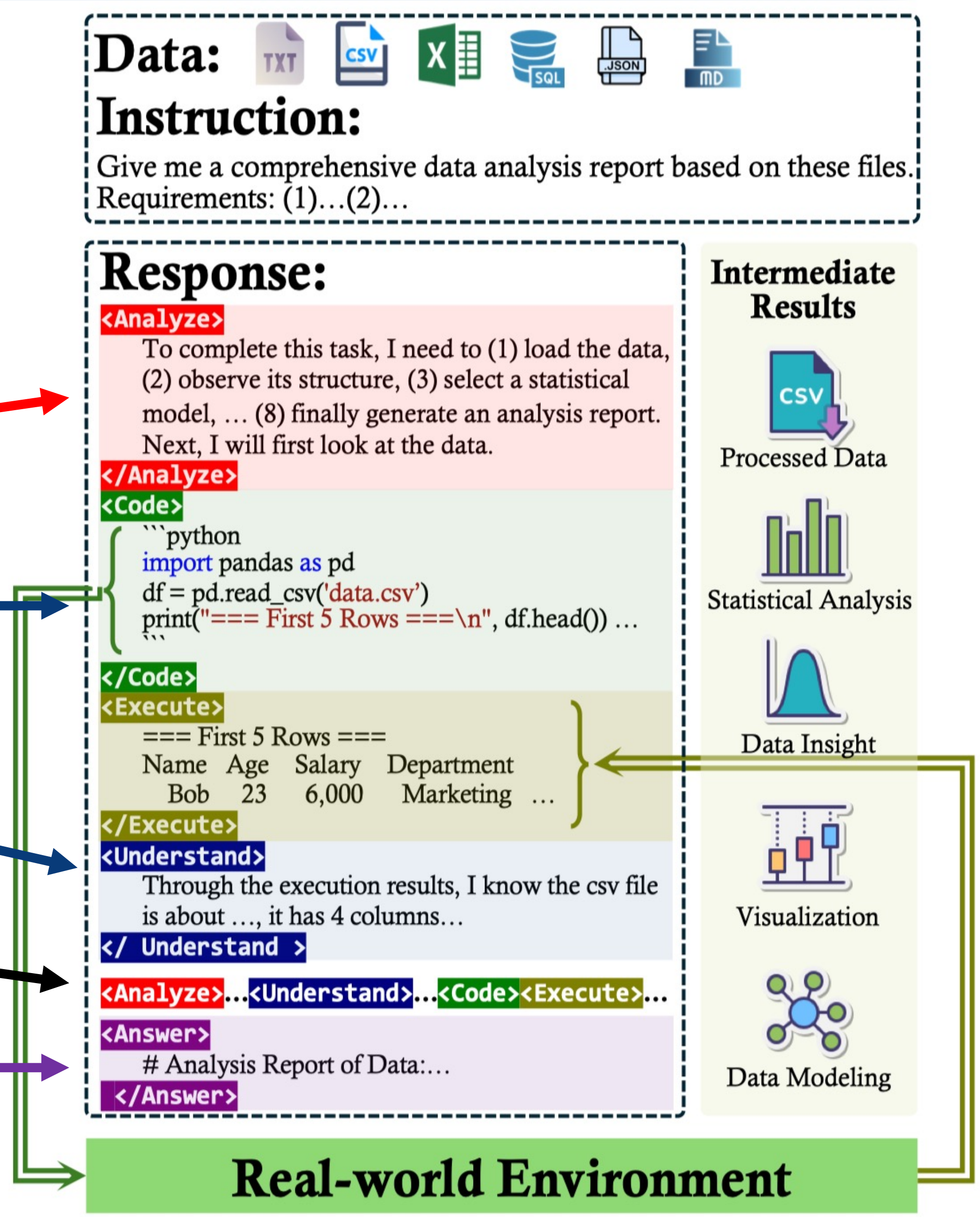


开源Code/Demo
3800+ stars

DeepAnalyze：自主数据科学大模型

引入若干action，使LLM能与数据交互

数据科学家	DeepAnalyze
完成数据科学任务的步骤：	
1. 分析任务，并规划步骤	<Analyze>
2. 在环境中读取数据	<Code>&<Execute>
3. 观察并理解数据	<Understand>
4. 通过代码完成数据分析、建模、可视化...重复直到完成任务	自动执行各action
5. 总结、呈现结果	<Answer>



DeepAnalyze：自主数据科学大模型

引入若干action，使LLM能与数据交互

数据科学家 | **DeepAnalyze**

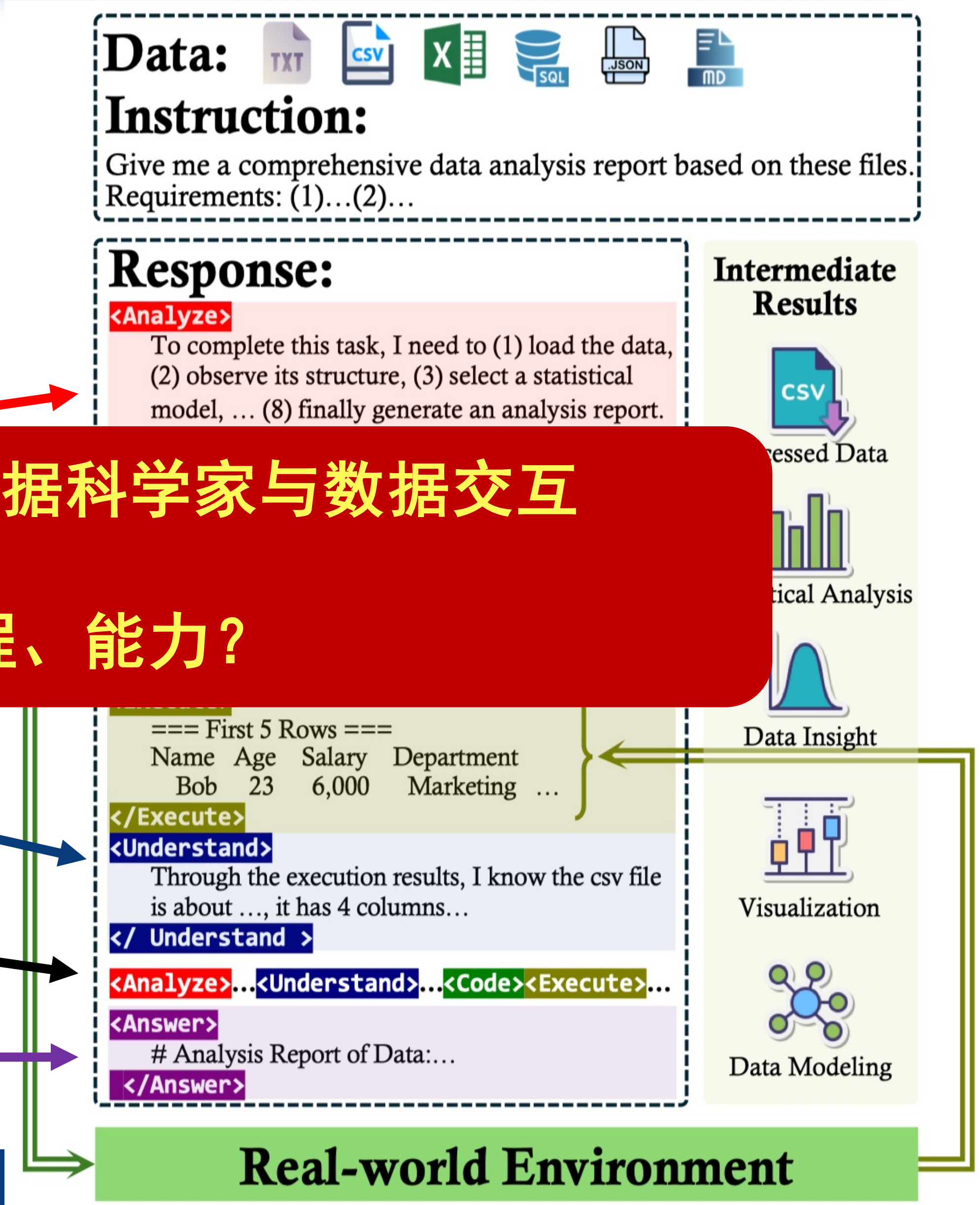
完成数据科学任务的步骤：

1.
2.
3.
4. 通过代码完成数据分析、建模、可视化…重复直到完成任务
5. 总结、呈现结果

<Understand>

自动执行各action

<Answer>



通过action定义，DeepAnalyze能模仿数据科学家与数据交互

⇓

如何学习数据科学思维、流程、能力？

DeepAnalyze：自主数据科学大模型

- Agentic Training：在真实环境中通过强化学习训练LLM
  - 在深度搜索等领域已经成功实践

- 在数据科学任务上的局限性：
  - 奖励稀疏：
    - 数据科学任务复杂 → LLM无法完成任务 → 正向奖励稀疏，LLM无法训练
  - 轨迹稀缺：
    - 完成任务轨迹空间大 → 缺少正确的推理/交互轨迹 → LLM陷入盲目试错

DeepAnalyze：自主数据科学大模型

- Agentic Training：在真实环境中通过强化学习训练LLM
  - 在深度搜索等领域已经成功实践

- 在数据科学任务上的局限性：
  - 奖励稀疏：
    课程式Agentic Training
    - 数据科学任务复杂 → LLM无法完成任务 → 正向奖励稀疏，LLM无法训练
  - 轨迹稀缺：
    Agentic 数据合成
    - 完成任务轨迹空间大 → 缺少正确的推理/交互轨迹 → LLM陷入盲目试错

DeepAnalyze：自主数据科学大模型

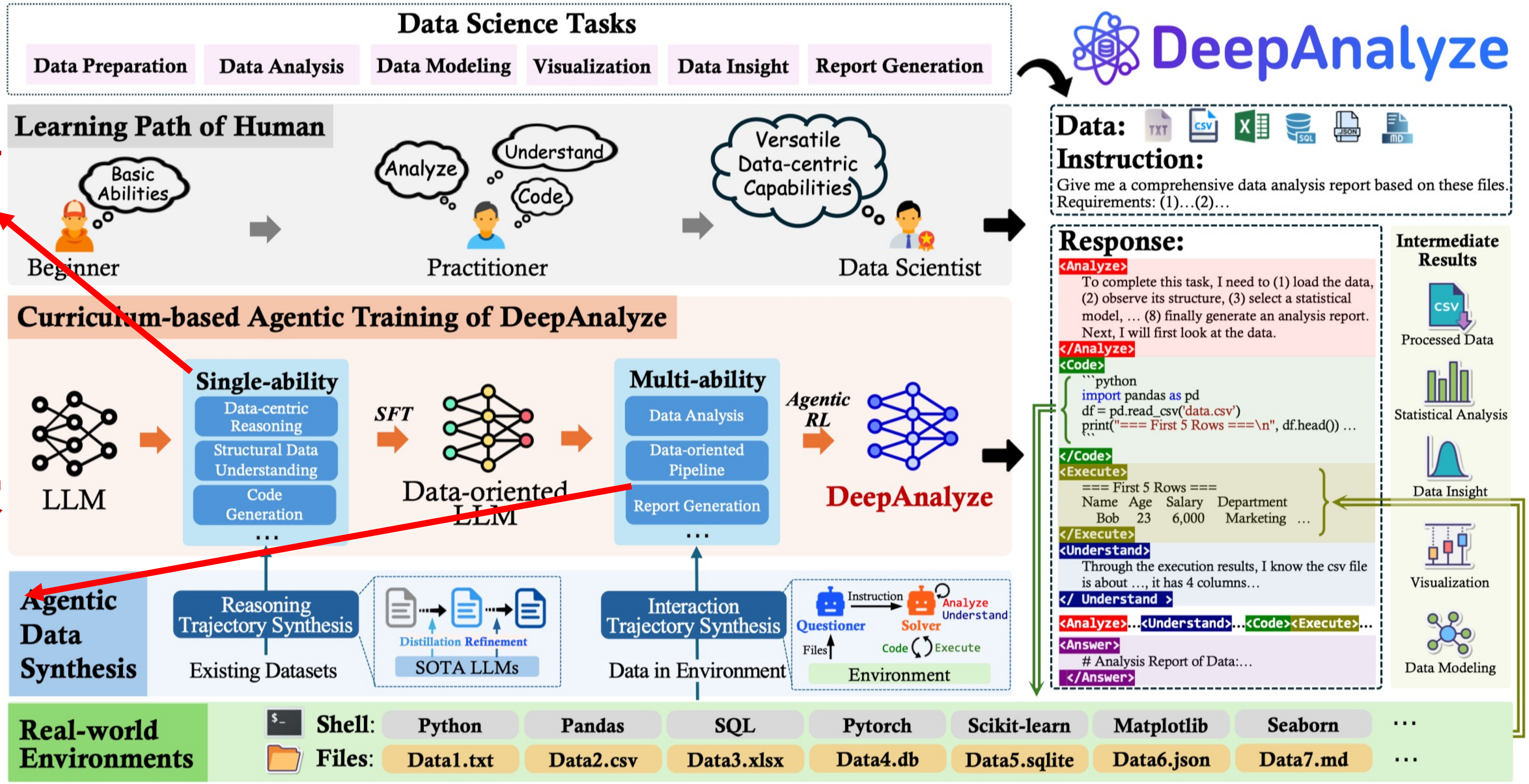
奖励稀疏：课程式Agentic Training　由单一能力到复合能力逐步学习

阶段1：初学者→从业者

- 学习推理、结构化数据理解、代码等单一能力

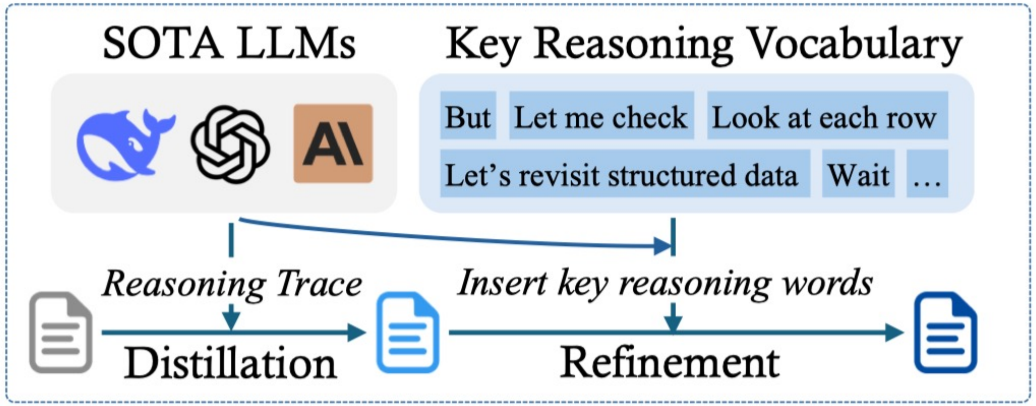
阶段2：从业者→科学家

- 学习多能力的复杂任务
- 真实环境中，自我进化

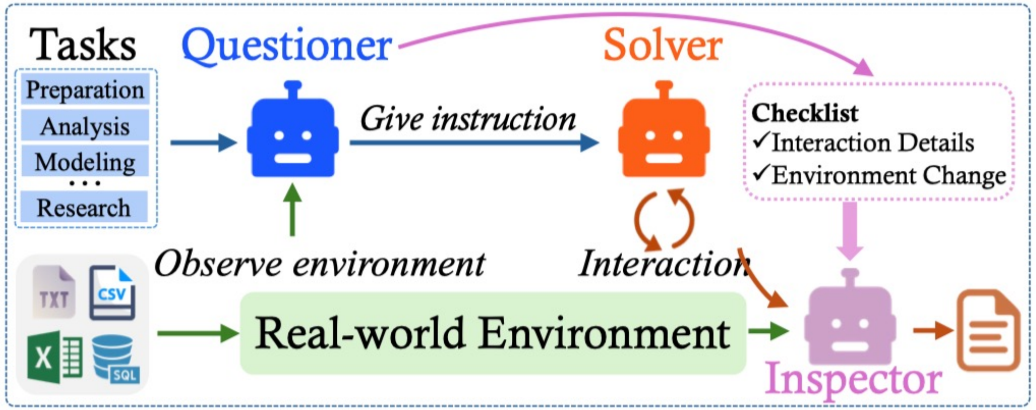


DeepAnalyze：自主数据科学大模型

轨迹稀缺：面向数据的轨迹合成框架 | **推理轨迹+交互轨迹，共500K数据**



(a) Reasoning Trajectory Synthesis



(b) Interaction Trajectory Synthesis

推理轨迹合成

- ✓ 蒸馏SOTA闭源模型
- ✓ 在推理轨迹中插入关键推理词汇（But/Wiat/Look at each row），增强推理轨迹对数据的关注

交互轨迹合成

- ✓ Questioner：根据数据提问题
- ✓ Solver：和环境交互完成任务
- ✓ Inspector：检测交互轨迹是否合法

DeepAnalyze：自主数据科学大模型

DeepAnalyze 在DataSciBench展现出全流程数据科学能力

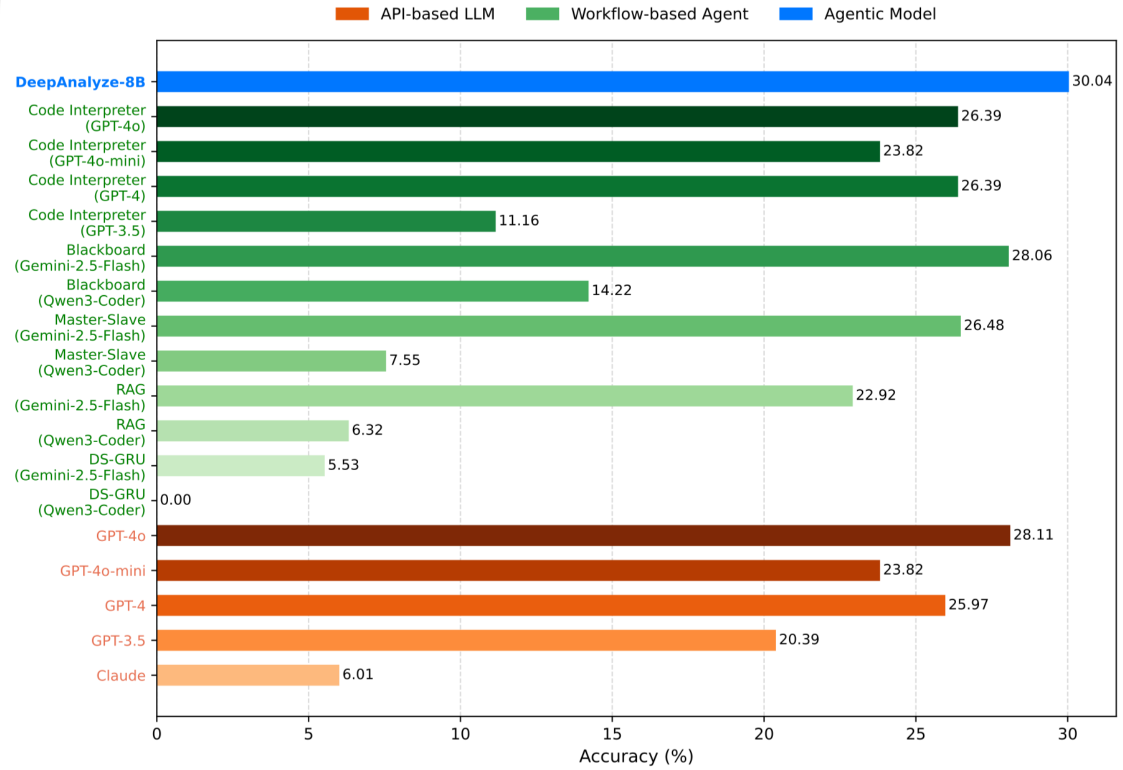
Models	Coarse-grained Metrics		Fine-grained Metrics						Score
	Success Rate	Completion Rate	VLM	F1: Data Preparation	F2: Plot Validity	F3: Data Exploration	F4: Data Visualization	F5: Data Modeling	
Close-Source API-Based Agent									
o1-mini	29.77	45.26	2.87	44.63	19.27	36.01	30.94	23.81	38.78
GPT-4o-mini	50.63	57.78	3.05	60.30	48.02	57.84	59.24	53.54	54.18
GPT-4o	**66.31**	**68.44**	**3.91**	**75.93**	**56.14**	**69.33**	**71.35**	**57.67**	**64.51**
GPT-4-Turbo	51.93	58.87	3.09	62.30	41.62	57.75	60.25	50.75	54.65
Claude-3-5-Sonnet	47.48	58.11	2.14	49.07	36.94	55.84	52.87	46.04	52.29
GLM-4-Flash	30.32	34.04	1.33	36.53	29.42	32.57	27.64	14.44	30.74
Open-Source LLM-based Agent									
Llama-3.1-8B-Instruct	24.73	33.89	1.29	38.24	18.25	21.98	22.89	25.85	29.69
Gemma-2-9B-it	7.07	11.00	1.06	26.16	16.90	23.81	18.11	17.15	12.66
GLM-4-9B-Chat	25.72	30.38	1.69	31.51	23.15	28.07	27.19	19.14	27.57
Qwen2.5-7B-Instruct	43.83	50.74	1.43	51.18	36.41	47.25	45.24	34.77	45.99
Qwen2-7B-Instruct	22.84	25.58	1.16	30.93	20.78	28.73	25.87	7.52	23.52
Yi-1.5-9B-Chat-16K	38.20	42.35	0.73	38.14	36.36	35.64	37.08	27.79	38.22
CodeLlama-13B-Instruct	10.49	14.64	0.04	11.67	11.34	9.43	14.43	5.15	12.64
CodeLlama-7B-Instruct	2.88	3.97	0.00	3.53	2.37	2.57	1.74	1.59	3.31
StarCoder2-15B	2.07	2.61	0.07	2.57	1.81	1.59	3.43	1.19	2.33
Deepseek-Coder-6.7B-instruct	37.03	41.62	1.93	43.49	34.57	46.36	46.49	18.09	38.45
Qwen2.5-Coder-7B-Instruct	45.18	53.11	1.48	51.58	43.21	43.87	42.50	**35.23**	47.67
Agentic Model									
DeepAnalyze-8B	**59.91**	**66.24**	**2.86**	**71.68**	**67.86**	**58.62**	**69.09**	33.33	**61.11**

F1:数据准备
F2:指令跟随
F3:数据分析
F4:数据可视化
F5:数据建模

8B模型媲美闭源模型

DeepAnalyze：自主数据科学大模型

DeepAnalyze 数据分析、数据建模能力



数据分析

Methods	LLM	Success (%)	Performance	Cost ($)
Workflow-based Agent				
AutoGen	Llama3-8b	5.41	1.55	0.00
	Llama3-70b	16.22	7.79	0.00
	GPT-3.5	8.11	6.02	0.41
	GPT-4	87.84	**45.52**	19.34
	GPT-4o	71.62	34.74	12.27
	GPT-4o-mini	22.97	11.24	0.10
Code Interpreter	GPT-3.5	16.22	6.52	2.74
	GPT-4	54.05	26.14	38.81
	GPT-4o	44.59	19.87	19.26
	GPT-4o-mini	39.19	16.90	2.70
Agentic Model				
DeepAnalyze-8B		**90.63**	39.41	**0.00**

数据建模

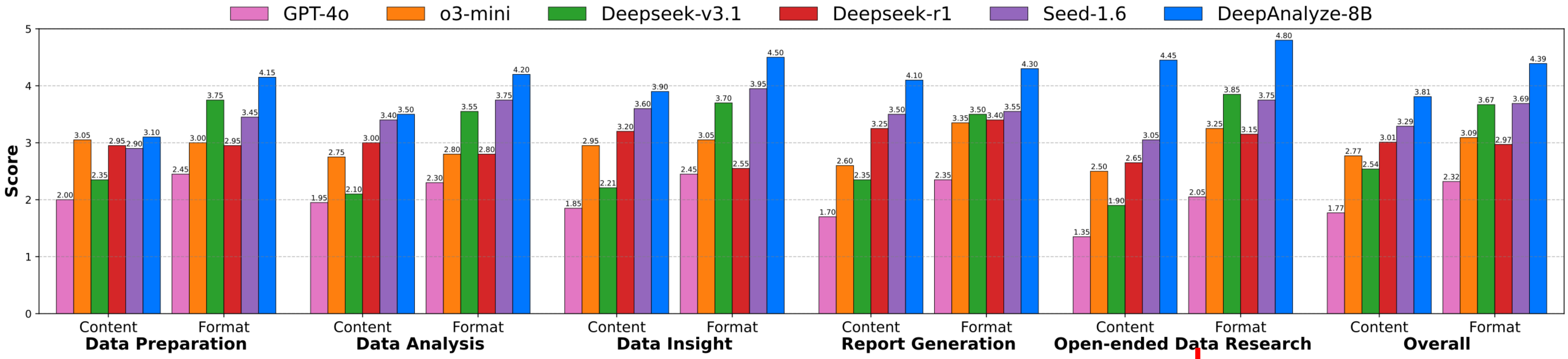
DeepAnalyze-8B优于workflow-based agent

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze：自主数据科学大模型

DeepAnalyze 深度数据研究能力

自动生成研究报告，从内容（content）、格式（Format）评分



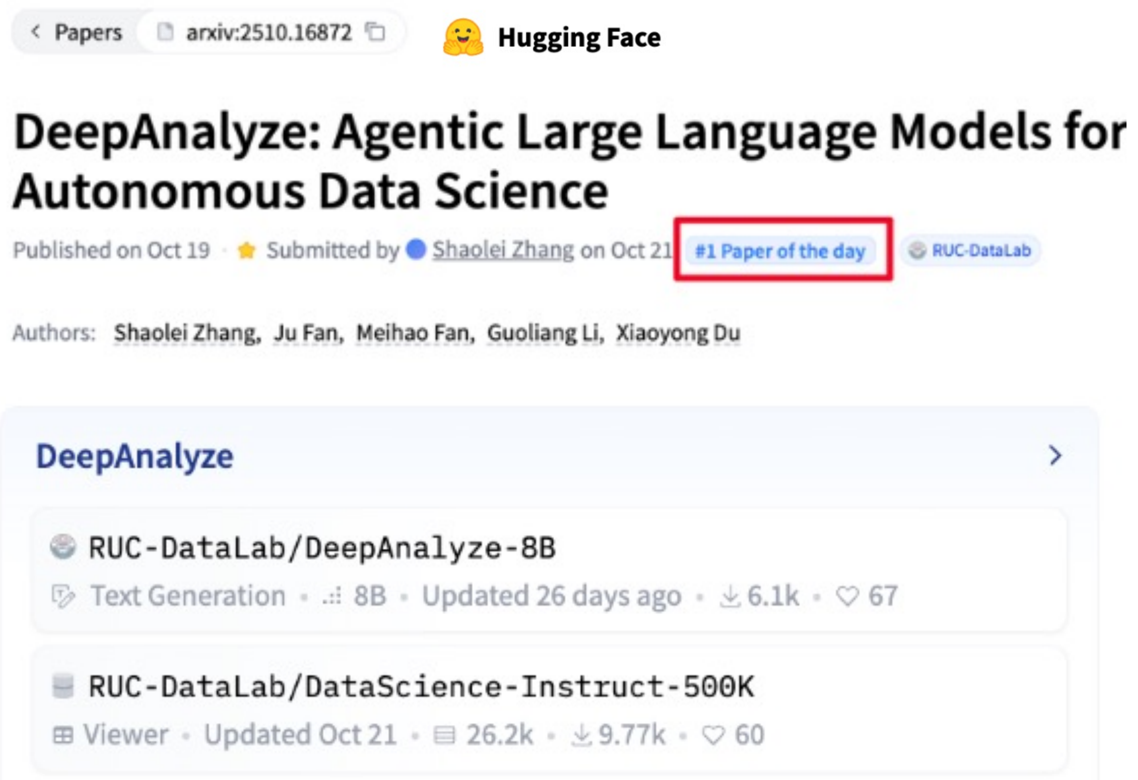
DeepAnalyze-8B优于闭源LLM（Tool Call）

闭源LLM未经训练，在指令中没有具体约束时，无法完成开放式数据研究

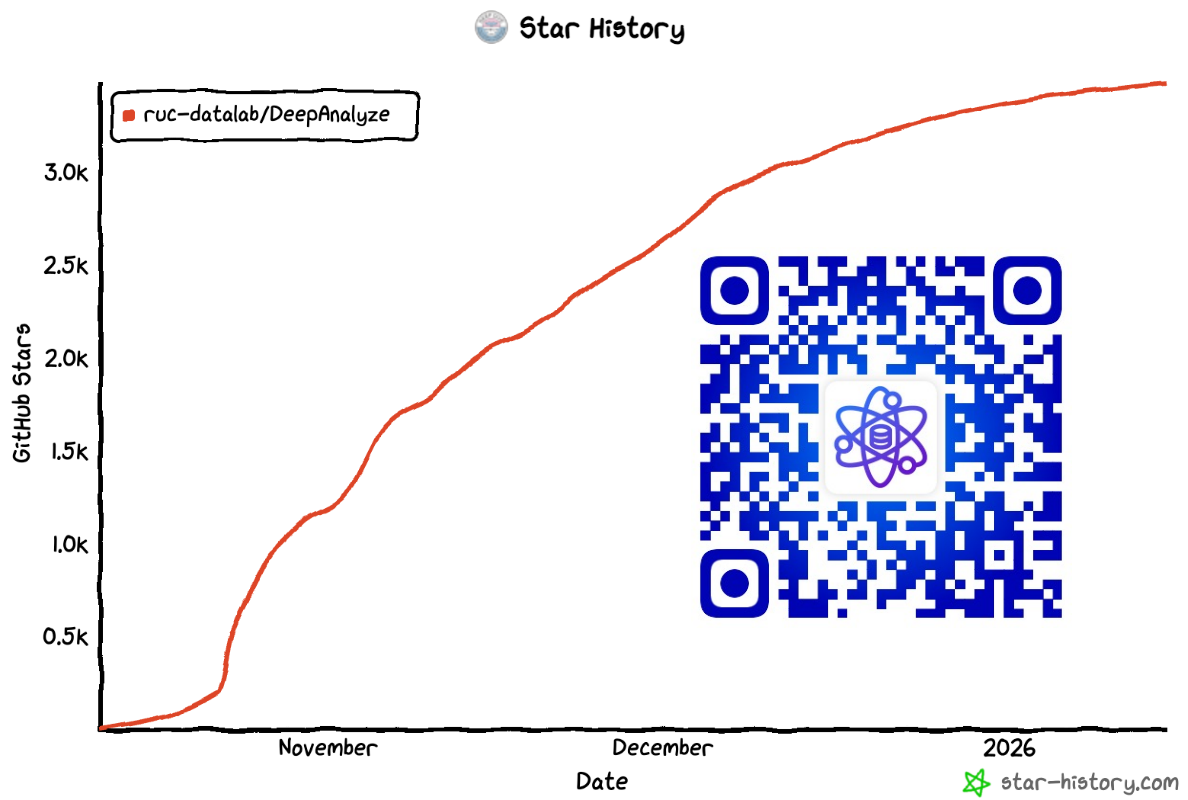
DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze：自主数据科学大模型

在HuggingFace上开源模型下载**2万+**，开源数据集下载**2万+**，曾在Huggingface日趋势榜**论文排名第一**



DeepAnalyze代码于2025年10月21日在GitHub上正式开源，已获得**3800+**个GitHub星标



DeepAnalyze引起社区广泛讨论

Brian Roemmele @BrianRoemmele · 13h

BOOM!

Powerful scientific research now performed by your local AI is here!

Garage researchers with no pedigree can now lift humanity to new height of discovery and we ain't asking permission.

I have been testing **DeepAnalyze**-8B, an open-source agentic language model and am

Show more

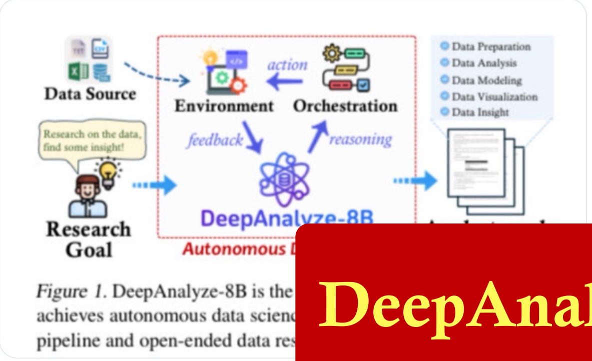


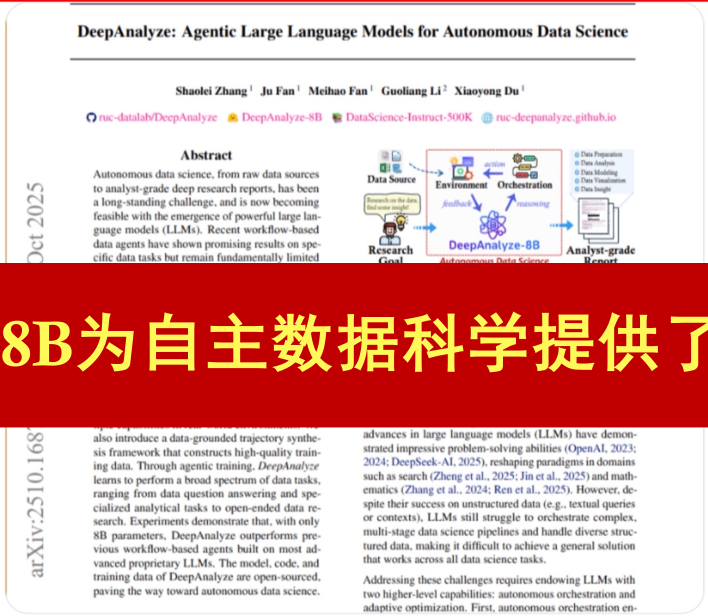
Figure 1. DeepAnalyze-8B is the ... achieves autonomous data scienc... pipeline and open-ended data res...

25 · 74

Dr Singularity @Dr_Singularity · 14h

Science is rapidly getting automated, and that means a 1000x acceleration is near.

DeepAnalyze-8B, the first agentic LLM designed for autonomous data science, capable of automatically completing the end to end pipeline from data sources to analyst grade deep research report.



DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Shaolei Zhang[1] Ju Fan[1] Meihao Fan[1] Guoliang Li[2] Xiaoyong Du[1]

ruc-datalab/DeepAnalyze · DeepAnalyze-8B · DataScience-Instruct-500K · ruc-deepanalyze.github.io

Abstract

Autonomous data science, from raw data sources to analyst-grade deep research reports, has been a long-standing challenge, and is now becoming feasible with the emergence of powerful large language models (LLMs). Recent workflow-based data agents have shown promising results on specific data tasks but remain fundamentally limited ...

... also introduce a data-grounded trajectory synthesis framework that constructs high-quality training data. Through agentic training, DeepAnalyze learns to perform a broad spectrum of data tasks, ranging from data question answering and specialized analytical tasks to open-ended data research. Experiments demonstrate that, with only 8B parameters, DeepAnalyze outperforms previous workflow-based agents built on most advanced proprietary LLMs. The model, code, and training data of DeepAnalyze are open-sourced, paving the way toward autonomous data science.

... advances in large language models (LLMs) have demonstrated impressive problem-solving abilities (OpenAI, 2023; 2024; DeepSeek-AI, 2025), reshaping paradigms in domains such as search (Zheng et al., 2025; Jin et al., 2025) and mathematics (Zhang et al., 2024; Ren et al., 2025). However, despite their success on unstructured data (e.g., textual queries or contexts), LLMs still struggle to orchestrate complex, multi-stage data science pipelines and handle diverse structured data, making it difficult to achieve a general solution that works across all data science tasks.

Addressing these challenges requires endowing LLMs with two higher-level capabilities: autonomous orchestration and adaptive optimization. First, autonomous orchestration en-...

arXiv:2510.168... Oct 2025

DeepAnalyze-8B为自主数据科学提供了一种解决方案

Rohan Paul @rohanpaul_ai · 23h

This paper builds an agentic LLM that can run the whole data science workflow by itself.

It is an 8B model that plans work, reads structured files, writes and runs code, checks results, and iterate...

26 · 77 · 480 · 30K

Shaolei Zhang @Vily_1998

Welcome everyone to use and submit issues & pull requests — we're very happy to collaborate with you all in contributing to DeepAnalyze.

Python Developer @Python_Dv · 16h

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze is the first agentic LLM for autonomous data science. It can autonomously complete a wide range of data-centric tasks without human intervention, supporting:...

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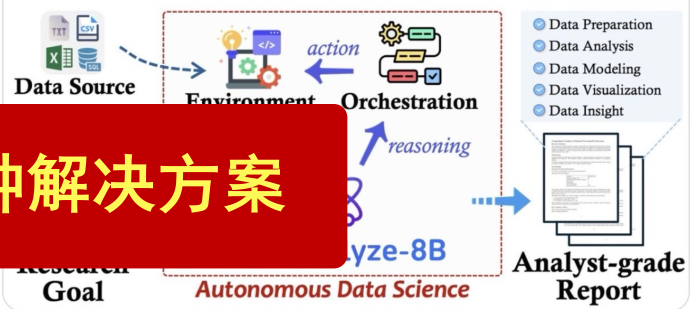

11:41 AM · Oct 22, 2025 · **28** Views

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Reply

社区反馈

DeepAnalyze在数据分析上显著优于通用模型上构建的智能体

光影织梦 @SunMousekomre · 20m

We tested it. The fine-tuned model has an accuracy rate that is 20-30% higher than the agent analysis we did based on the general model.

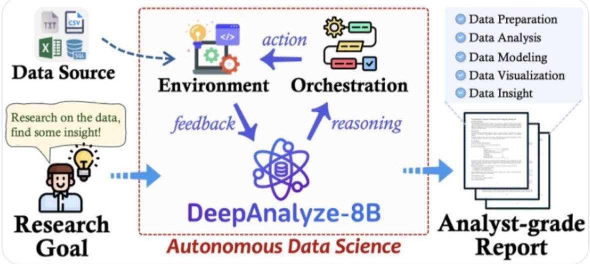
1 · 4

Gorden Sun @Gorden_Sun · Oct 21

DeepAnalyze：数据分析智能体

能自主完成一系列数据任务，包括：准备、分析、建模、可视化、生成报告。

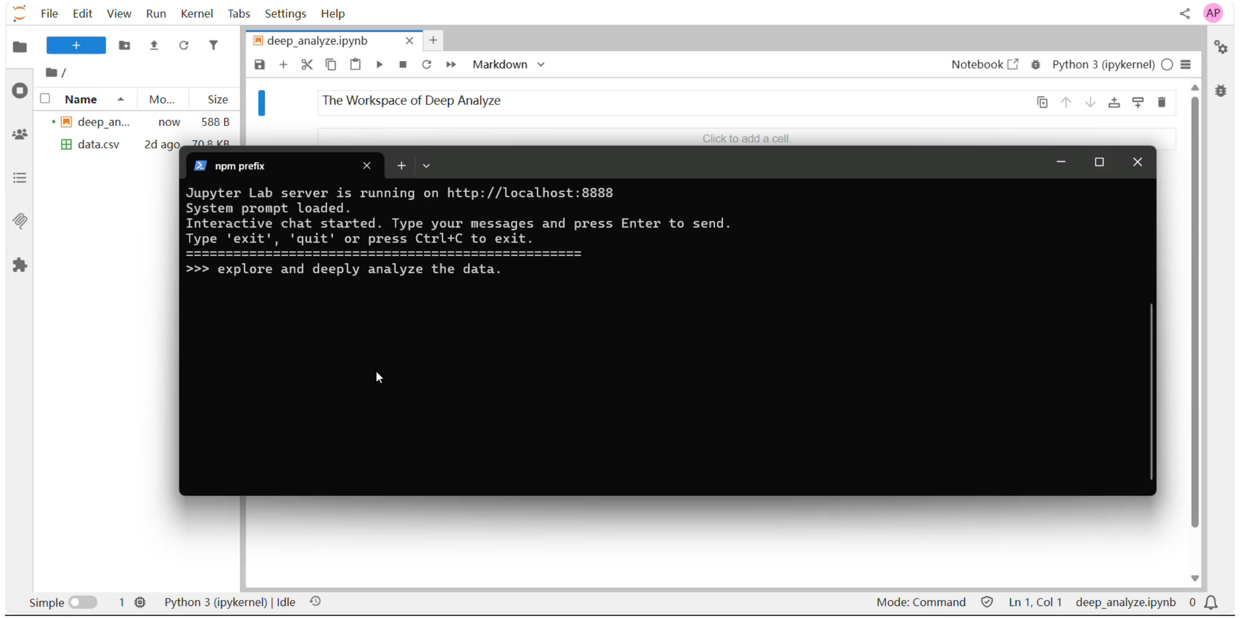
Github：github.com/ruc-datalab/De...

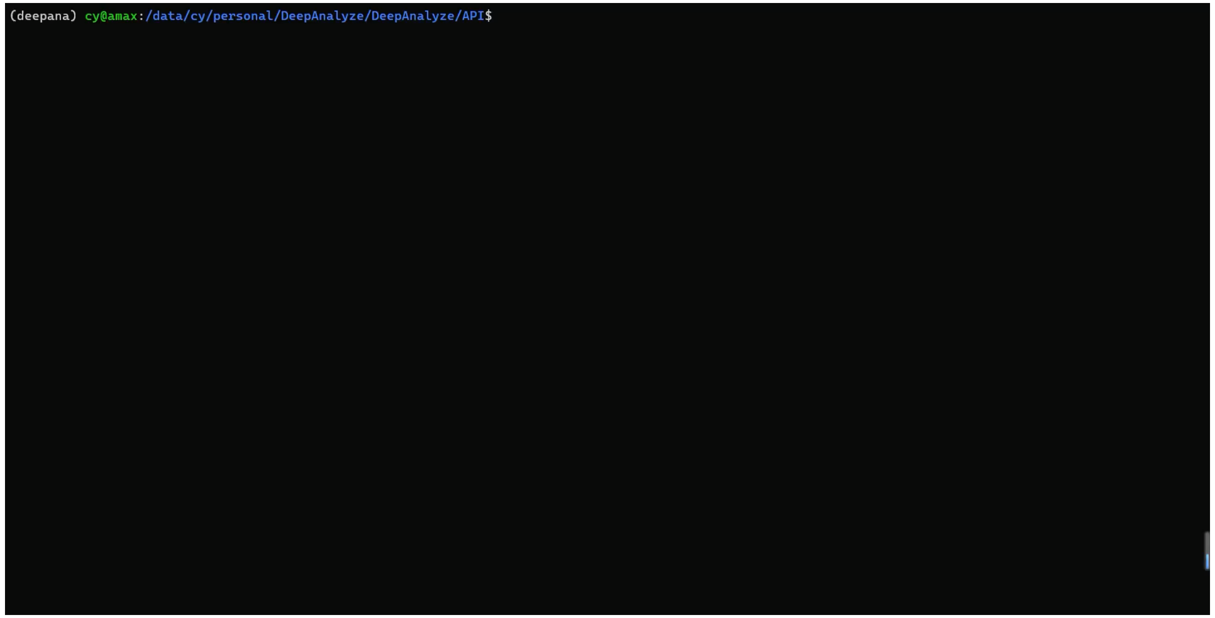


3 · 69 · 219 · 16K

DeepAnalyze在社区实践中初具成效

人机协同方式 通过WebUI、JupyterUI、CLI模式交互





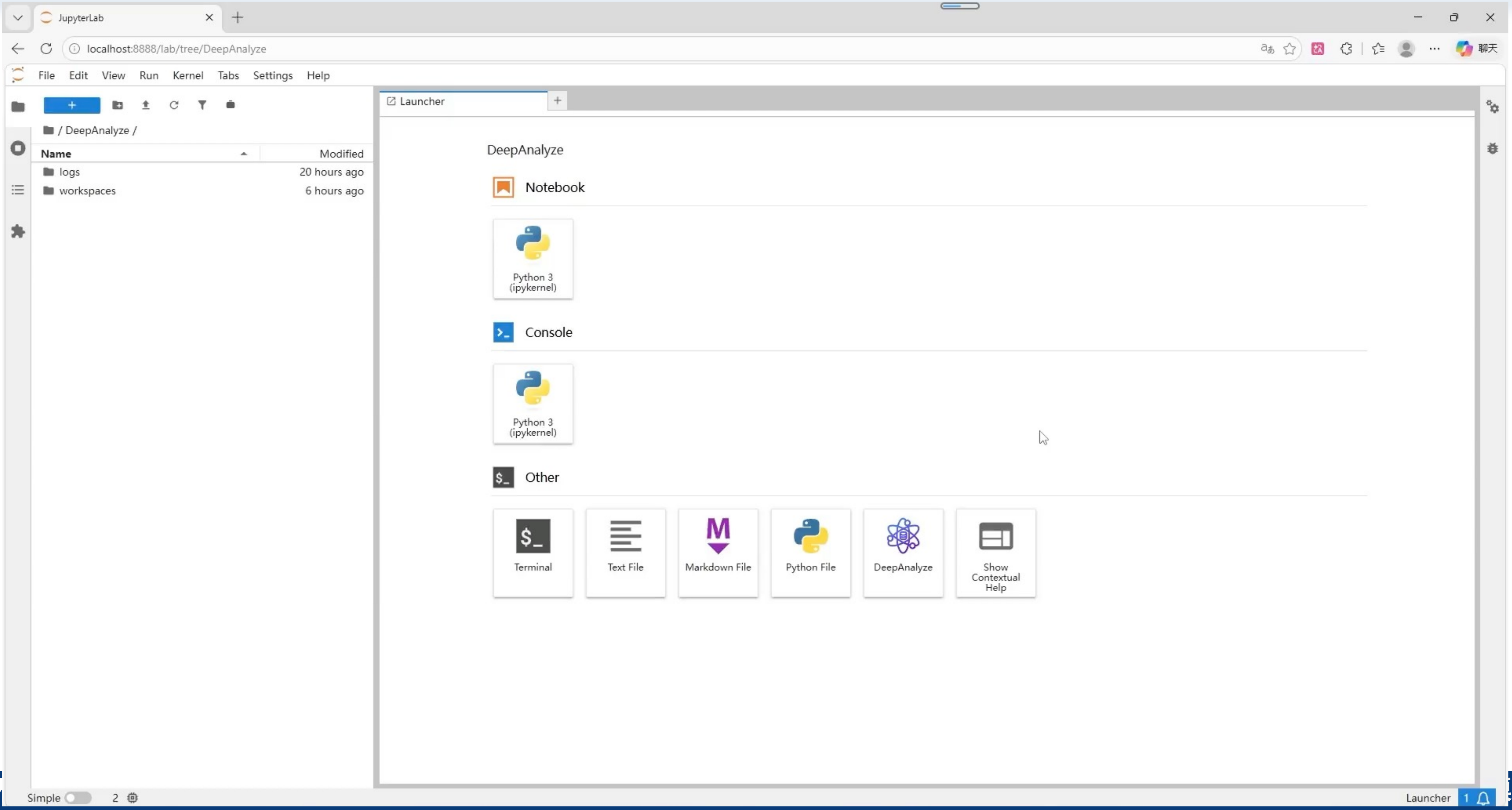
JupyterUI

通过自然语言控制Jupyter notebook对数据进行分析

CLI

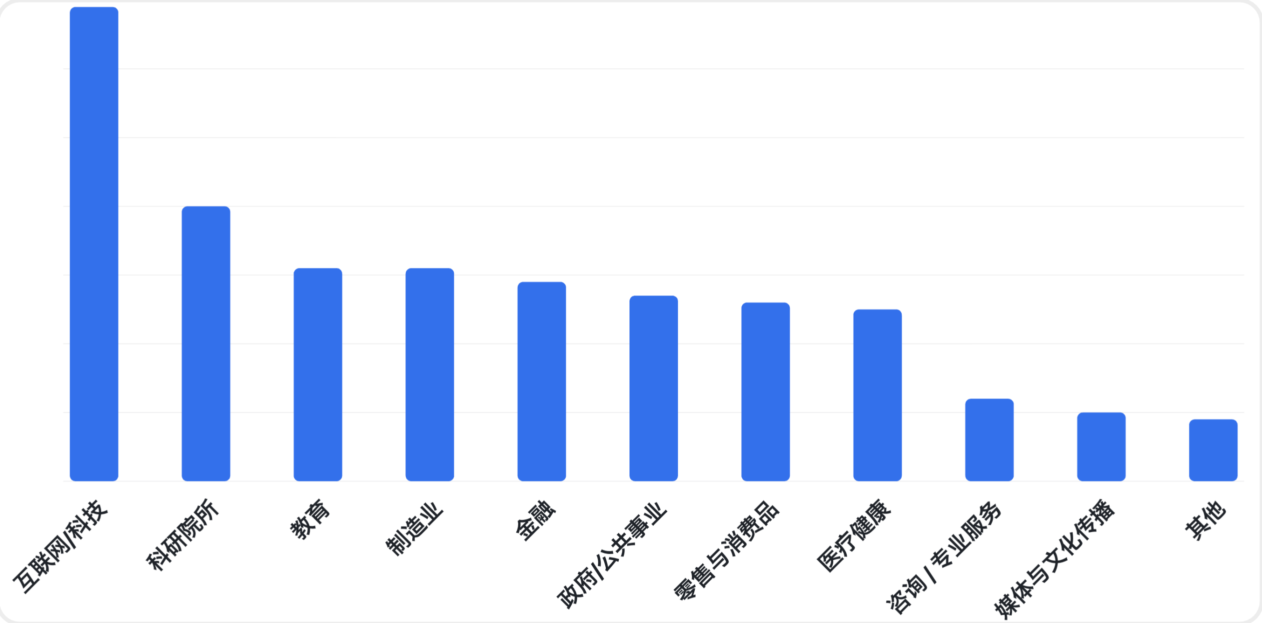
通过命令行交互对数据进行分析

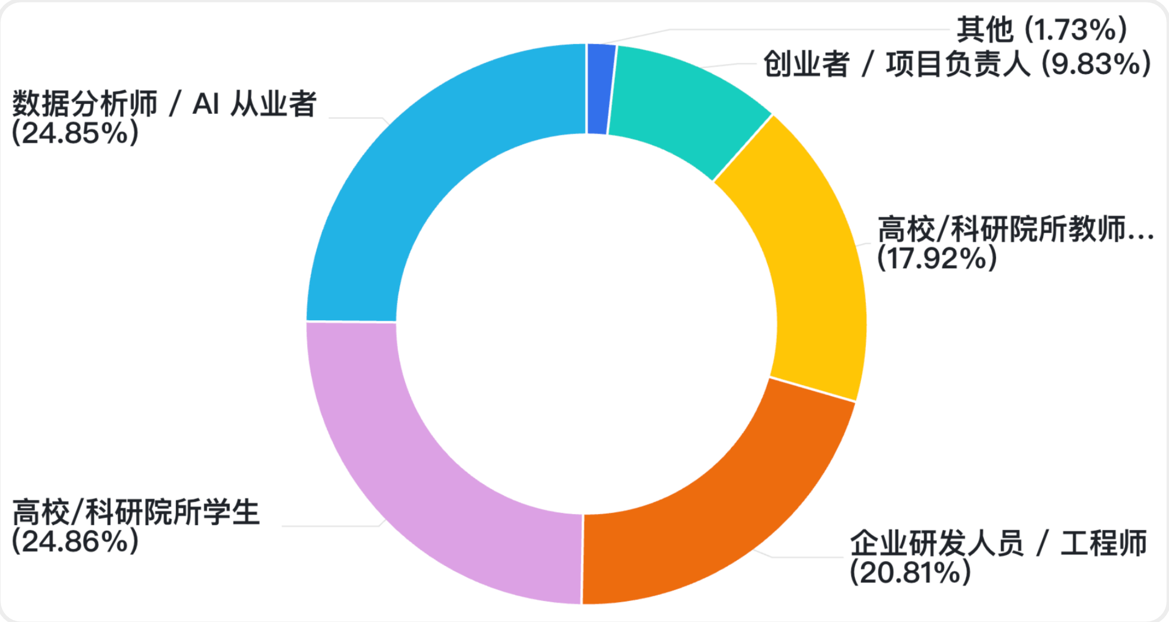
Vibe Analyzing with DeepAnalyze



DeepAnalyze在社区实践中初具成效

DeepAnalyze × 和鲸 收到400+来自院校/企业的内测申请





跨领域应用潜力巨大

来自互联网、教育、金融、制造、医疗、政府、零售、咨询各领域

深度覆盖“研”与“用”群体

“研”：高校/科研院所教师、学生、研究员

“用”：数据分析师、AI从业者、创业者

总结

DeepAnalyze 首个面向数据科学的Agentic大模型

- 课程式Agentic训练 + Agentic数据合成
- DeepAnalyze作为数据科学领域的基座模型 → 直接使用、特定领域微调、搭建复杂工作流

DeepAnalyze
交流群

DeepAnalyze讨论群

张绍磊
北京 东城

联系方式

谢 谢 大 家

张绍磊

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https://zhangshaolei1998.github.io/

联系方式

个人主页